

Anti Viral agents

Frehiwot Beyene

- Antiviral agents are drugs that control virus infections.
- Unlike antibacterial drugs, which may cover a wide range of pathogens, antiviral agents tend to be narrow in spectrum, and have limited efficacy.
- As a class, the antiviral drugs are not usually curative, and must be used either prophylactically or early in the development of an infection

Strategies for Antiviral Therapy

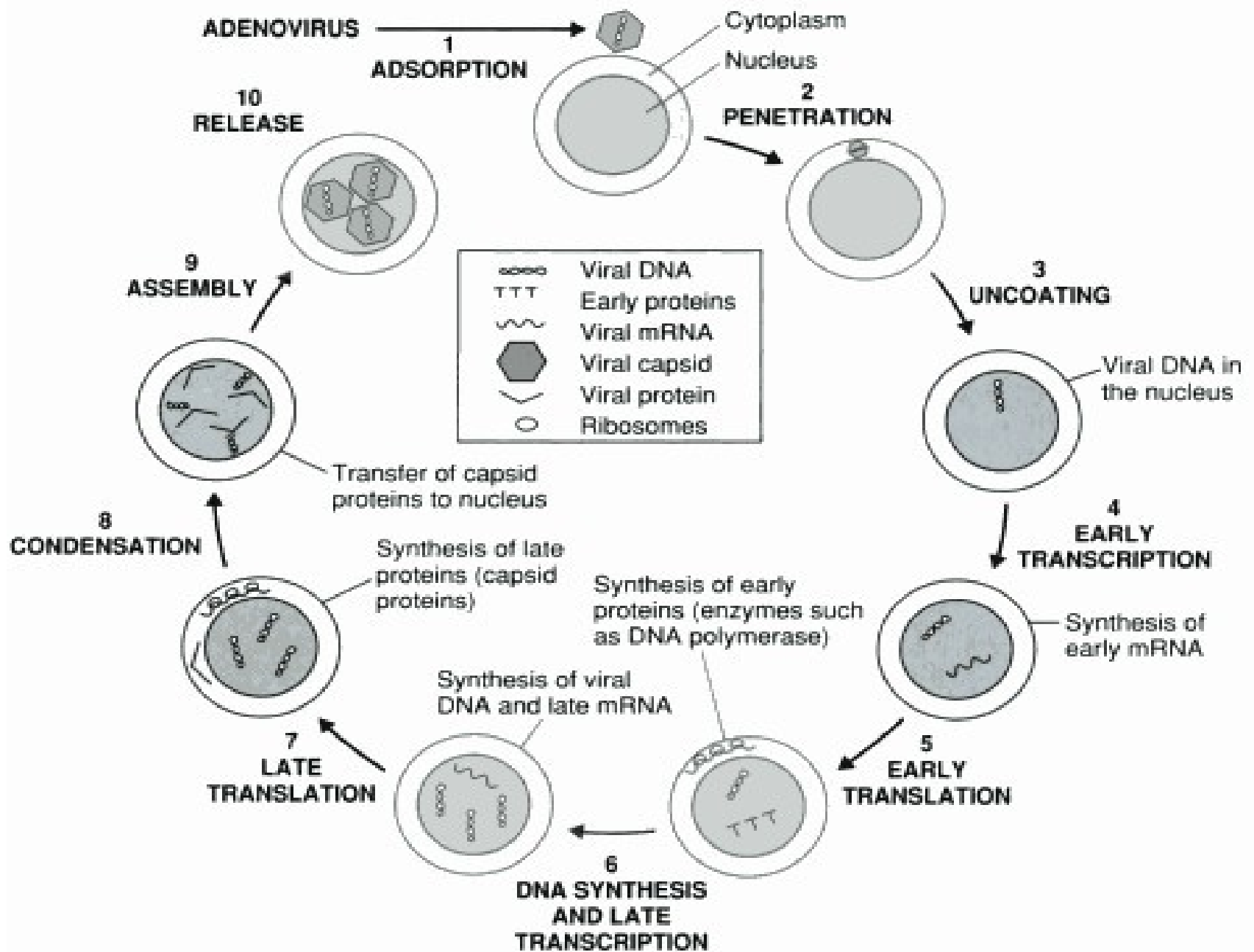
- Any stage of viral replication with the following requirements can be a target for antiviral drugs.
 - The stage targeted is essential for virus replication.
 - The therapeutic agent is active against the virus while having "acceptable toxicity" to the host organism

Stages in Virus Replication which are Possible

Targets for

Chemotherapeutic Agents

- Attachment to host cell.
- Uncoating - (Amantadine).
- Replication of viral RNA or DNA - (Nucleoside analogues).
- Synthesis of viral mRNA - (Interferon)
- Translation of mRNA - (Interferon).
- Maturation of new virus proteins (Protease inhibitors).-
- Budding, release.



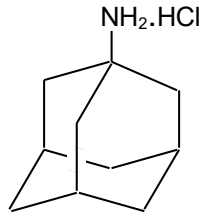
Classification of Antivirals

- Antivirals effective against influenza virus.
- Antivirals effective against herpes virus.
- Antivirals effective against respiratory syncytial virus.
- Antivirals effective against hepatitis C and B.
- Antivirals effective against HIV virus.

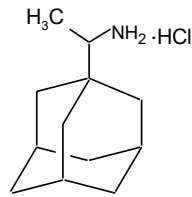
Antiviral Agents Effective Against Influenza Viruses

A. Agents Effective Against Influenza Virus Type A

- Adamantane Derivatives
 - i. Amantadine HCl (Adamine®)
 - ii. Rimantadine HCl (Flumadine®)



1-Adamantanamine
Hydrochloride



α-Methyl-adamantanemethylamine
hydrochloride

- The clinical usefulness of amantadine and rimantadine is limited to the prevention and treatment of *influenza A virus* infections.
- Amantadine is excreted by the kidneys unchanged via glomerular filtration and tubular secretion while rimantadine is metabolized by the liver (75%) and is also excreted by the kidneys.
- Rimantadine also has a longer duration of action.

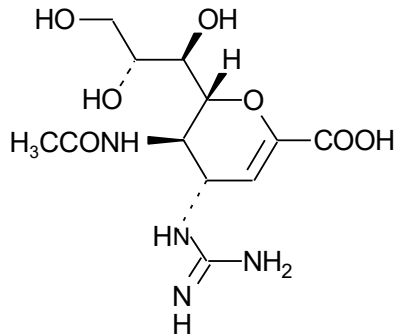
MOA

- The two drugs inhibit the early stages of viral replication, blocking the uncoating of the viral genome and the transfer of nucleic acid into the host cells.
- Amantadine inhibits also virus particle penetration into the host cells.

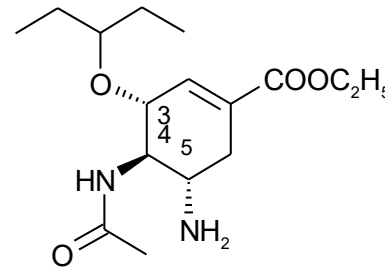
Agents Effective against Virus Influenza Types A and B

Neuraminidase Inhibitors (Sialic Acid Analogs)

Zanamivir

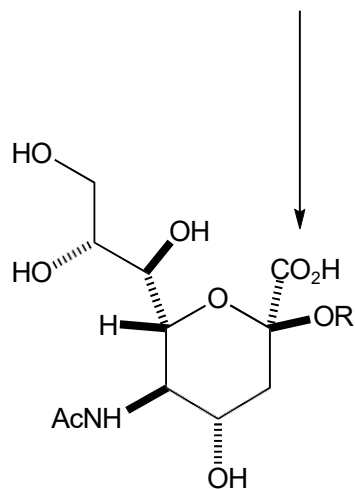


Oseltamivir



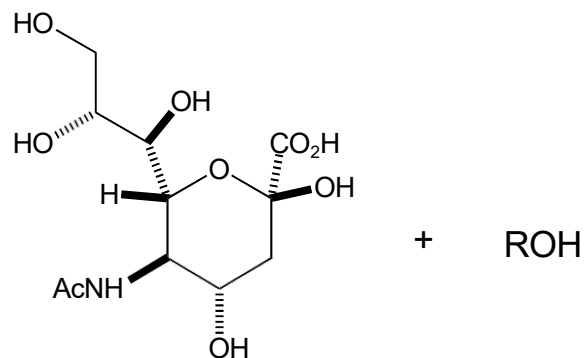
- The neuraminidase inhibitors, zanamivir and oseltamivir, are chemically related drugs that have activity against both influenza A and B viruses.
- Zanamivir is an orally inhaled powdered drug that is approved for treatment of influenza in persons aged 7 years and older.
- Zanamivir is not approved for chemoprophylaxis of influenza.
- Oseltamivir is an ethyl ester prodrug requiring ester hydrolysis for conversion to the active form, oseltamivir carboxylate
- Oseltamivir is an orally administered capsule or oral suspension that is approved for treatment of influenza in persons aged 1 year and older.
 - It is also approved for chemoprophylaxis of influenza in persons aged 13 years and older.

Negatively charged at neutral pH



Neuraminidase

H₂O



Sialic Acid
(also: N-acetylneuraminic acid)

Mechanism of Action

- Influenza viruses are surrounded by a protein coat and a lipid envelope.
 - Embedded in the lipid membrane there are two surface glycoproteins:
 - Hemagglutinin (HA) is an enzyme important for binding viruses to target cell receptor via a terminal sialic acid residue.
 - Neuraminidase (NA) an enzyme involved in various activation of influenza viruses.
 - NA is found in both influenza A and B and is thought to be involved in catalytically cleaving glycosidic bonds between terminal sialic acid and an adjacent sugar.
 - The cleavage of sialic acid bonds facilitates the spread of viruses and as a result increases the infectiveness or pathogenicity of the virus.
 - NA inhibitors interfere with the spread of the infection by blocking sialic acid bonds cleavage.

Antiviral Agents Effective Against Herpes Virus

The family of herpes virus includes:

- Herpes simplex types 1 and 2 (HSV1, HSV2), Varicella Zoster (VZV) and Epstein Bar (EBV).
- Cytomegalovirus (CMV).

A) Agents Effective Against HSV1 and 2, VZV and EBV

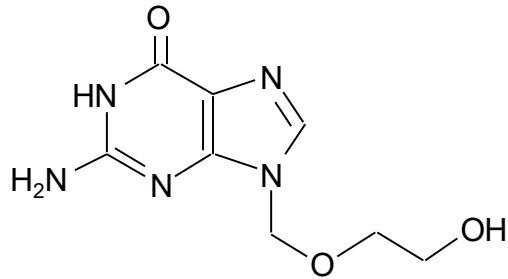
- Drugs in this class are Nucleotide analogues with sugar modification.
- They act as chain terminators and DNA polymerase inhibitors.

i. Acyclovir (Zovirax)

ii.

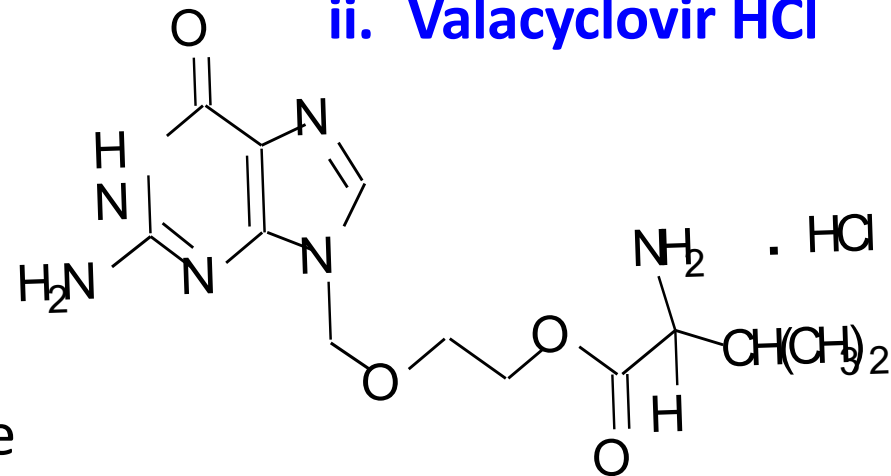
Valacyclovir HCl

i. Acyclovir (Zovirax)



9-[(2-Hydroxyethoxy)methyl]guanine

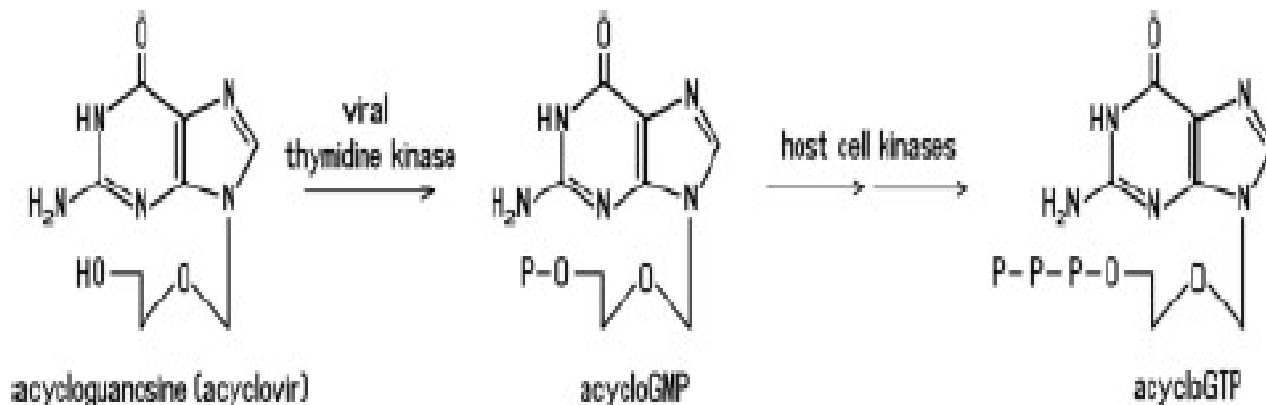
ii. Valacyclovir HCl



- Acyclovir is a synthetic analog of the normal metabolite 2'-Deoxyguanosine.
- Valacyclovir is a valine ester of acyclovir that is well absorbed. Its bioavailability is 2-5 greater than acyclovir.
- So low dose is possible at less frequent intervals.
- It is used for the treatment and suppression of genital herpes infection.

MOA

- Viral TK phosphorylates acyclovir to acyclovir monophosphate.
- Host cellular enzymes further phosphorylate acyclovir monophosphate to acyclovir triphosphate, the active form.
- Acyclovir triphosphate competes with the natural nucleoside, **deoxyguanosine triphosphate (dGTP)**, for incorporation into viral DNA by the viral DNA polymerase.
- Chain termination occurs because the drug triphosphate lacks the 3-hydroxy group of cyclic sugar.

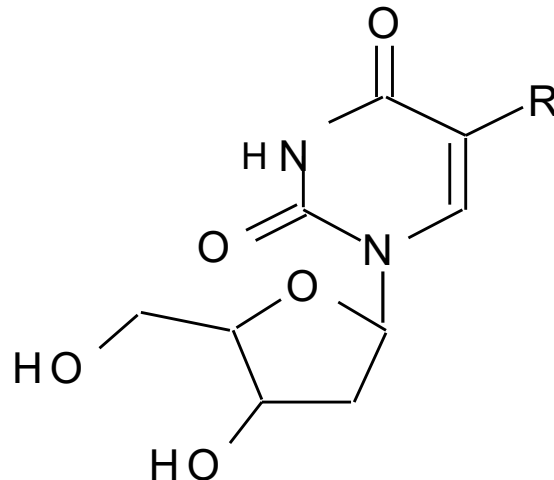


Selectivity of Action

- These drugs reduce DNA synthesis in virus-infected cells without affecting the active replication of uninfected cells because:
 - Viral thymidine kinase phosphorylates the drug 100 times faster than host cells (differential toxicity).
 - Acyclovir triphosphate competitively inhibits viral DNA polymerases, and to a much smaller extent cellular DNA polymerases.

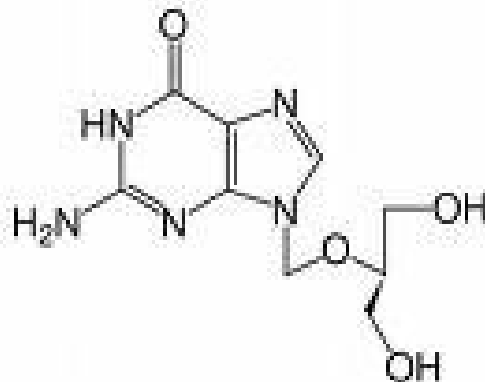
Other Drugs Used as Topical Applications for Herpes

- Tifluridine (R=CF₃), 2'-deoxy-5-(trifluoromethyl) uridine
- Idoxuridine (R = I), 2'-deoxy-5-iodouridine
 - These are pyrimidine analogs with base modification
 - Are only suitable for topical use.



Agents Effective Against Cytomegalovirus (CMV)

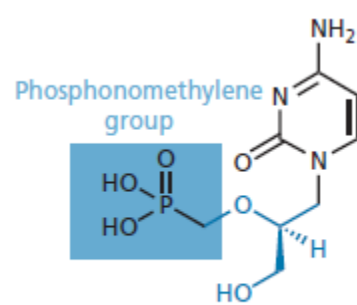
- Ganciclovir 9-[1,3-Dihydroxy-2-propoxymethyl]guanine
- Ganciclovir is a deoxyguanosine analog that differs from acyclovir in having an additional hydroxymethyl group on the acyclic side chain.
- It is the preferred drug for treating (CMV) infections in patients with acquired immune deficiency syndrome (AIDS) or other immunodeficiencies.
- It has very poor oral bioavailability (3 percent), and, therefore, mostly given intravenously.



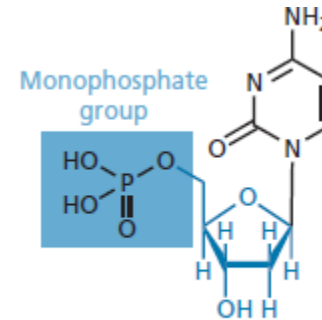
MOA

- It is an acyclic guanine nucleoside analog and is activated to the nucleoside triphosphate by CMV-specific phosphotransferase
- CMV is deficient in thymidine kinase this is why acyclovir has no activity against CMV.
- Selectivity is achieved because the viral polymerase has 30 times greater affinity for ganciclovir than the host enzyme.

Cidofovir –active against the virus with no thymidine kinase activation



Cidofovir

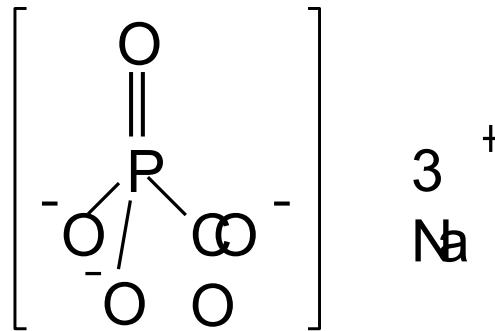


Deoxycytidine monophosphate

Foscarnet (Foscavir®)

Foscarnet

- It is trisodium phosphonoformate
- It is the drug used in the treatment of CMV infections, particularly CMV retinitis, in immunocompromised patients.



Mechanism of action

- It is an Inorganic pyrophosphate analog that works by reversibly blocking the pyrophosphate binding site on viral DNA polymerase, thereby terminating chain elongation.
- Unlike nucleoside analogs, foscarnet does not need to be activated by cellular or viral kinases.

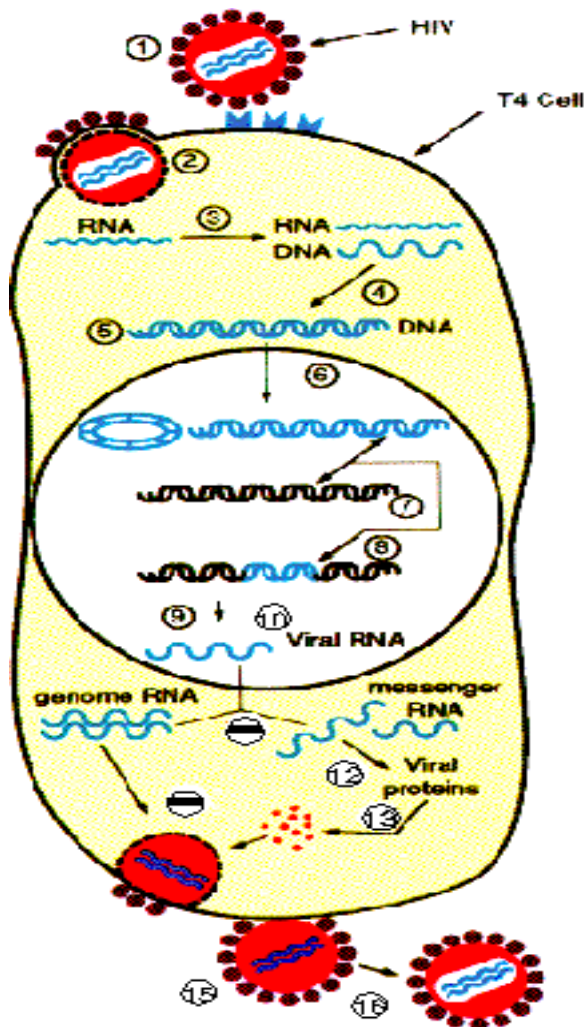
Anti-HIV Agents

- Billions of dollars have been spent on research and development of anti-AIDS drugs, the disease remains uncontrolled.
- Over 30 million people infected worldwide with HIV.
- Therapy of HIV is complicated by the ability of the virus to mutate leading to rapid drug resistance.
- Although a number of substances with *in vitro* anti-HIV activity have been described, only a few drugs exhibit anti-HIV activity *in vivo* at tolerable toxicities.

HIV target

- Cells containing more number of CD4 cells
- T4-lymphocyte (also called the "T-helper cell"), a kind of white blood cell that has lots of CD4 receptors.
- The T4-cell is responsible for warning your immune system that there are invaders in the system

Life cycle of HIV



- Step 1: Attachment of virus at the CD4 receptor and chemokine co-receptors CXCR4 or CCR5
- Step 2: viral fusion and uncoating
- Steps 3-5: Reverse transcriptase makes a single DNA copy of the viral RNA and then makes another to form a double stranded viral DNA
- Step 6: migration to nucleus
- Steps 7-8: Integration of the viral DNA into cellular DNA by the enzyme integrase
- Steps 9-11: Transcription and RNA processing
- Steps 12-13: Protein synthesis
- Step 14: protease cleaves polypeptides into functional HIV proteins and the virion assembles
- Step 15: virion budding
- Step 16: Virion maturation

- The most challenging problem associated with HIV therapy is the development of Resistance
- The combination therapy approach has been broadly practiced in HIV treatment as a strategy to avoid such problem development of drug resistance.
- Current efforts to develop new anti-HIV drugs are now directed more toward developing inhibitors for other steps in the virus life cycle such as
 - viral entry into the T-cell,
 - its integration with the host DNA, or
 - its assembly into daughter particles.
- These proposed sites of intervention may provide additional options to maintain long-term suppression of the virus replication

Inhibitors of Viral Entry

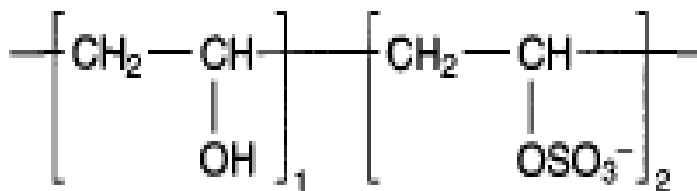
HIV Vaccines

- The most ideal approach to inhibit HIV binding to the T-cells is to develop a vaccine that can neutralize the virus circulation.
- Several attempts to develop antibodies to HIV as blocker for the viral binding to prevent infection have been reported.
- However, because of the nature of the disease, people infected with HIV develop only low titers of neutralizing antibodies and that presents a problem for vaccine development.

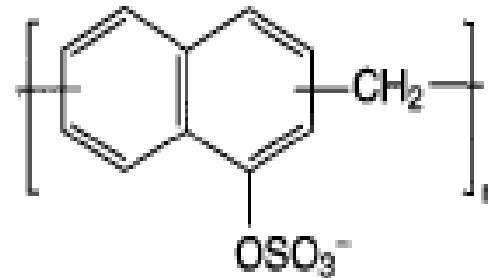
Inhibitors of Viral entry...

A. polyanionic compounds

- Several polyanionic compounds of natural and synthetic sources have been reported to inhibit gp120 binding to the cell membrane.
- Among these anionic compounds are polysulfates, polysulfonates, and polycarboxylates.
- All are believed to exert their antiviral activity through inhibiting gp120 binding to the CD4 receptor



PVAS

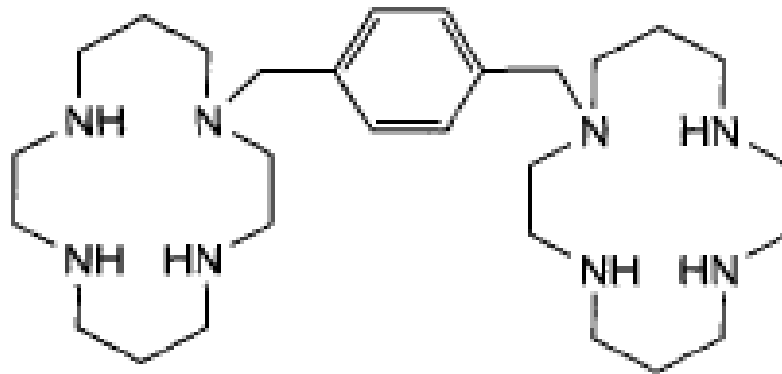


Naphthalene sulfonate

Inhibitors of viral entry...

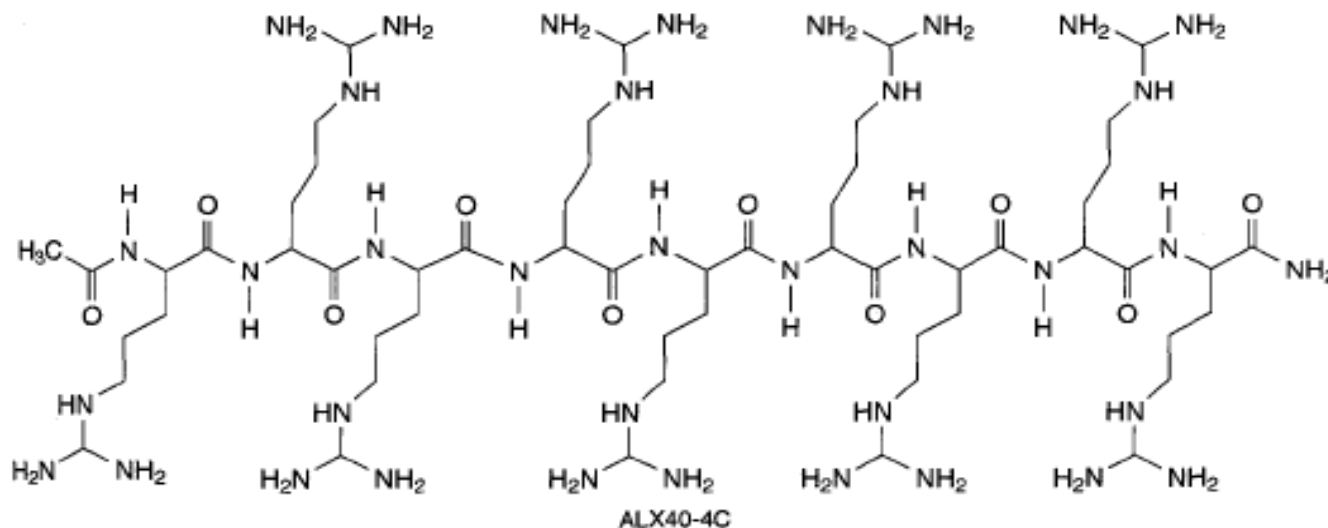
B. Inhibitors of Gp120 Binding to the T-cell Coreceptors

- CXCR4 and CCR5 are Gp120 Coreceptors which are found on Tcell membrane surface
- The most interesting member of Gp120-CXCR4 Binding Inhibitor is the compound bicyclam (AMD3100)

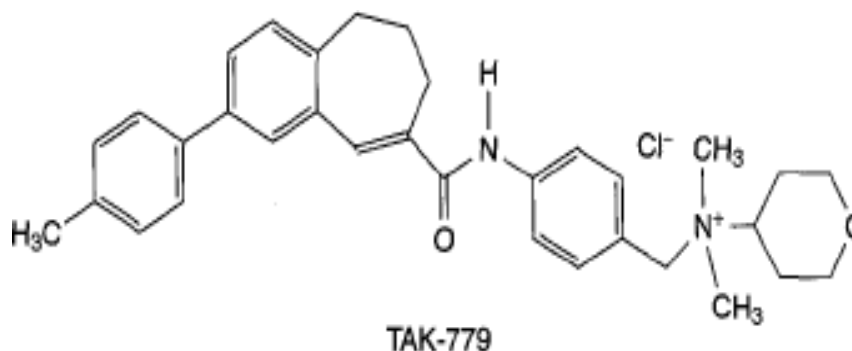


AMD3100

Additional preclinical derivatives include the polypeptide

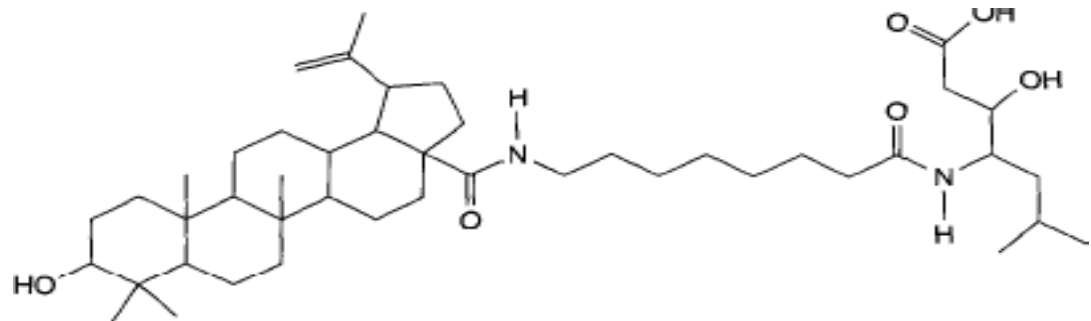


TAK-779 is a potent Gp120-CCR5 Binding Inhibitor which is under clinical trial



Inhibitors of Viral Fusion

- The fusion of the virus envelope with the Tcell membrane commences after the binding of gp120 to the coreceptors CXCR4 and CCR5.
- This binding triggers a spring-loaded action of the glycoprotein gp41, which is normally covered by the larger gp120.
- The gp41 anchors itself to the T-cell membrane through the hairpin structures HR1 and HR2
- The betulinic acid derivative RPR103611 represent a nonpeptide inhibitor for gp41 fusion.



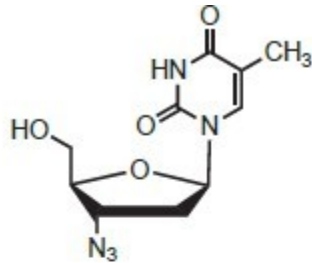
RPR103611

Currently available Anti Retroviral Agents are classified into three groups

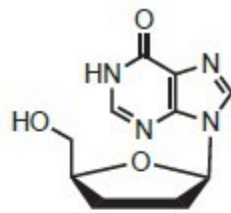
- Nucleoside analogs reverse transcriptase inhibitors (NRTI).
- Non-nucleoside analogs reverse transcriptase inhibitors (NNRTI).
- HIV Protease inhibitors.

Nucleoside Reverse Transcriptase Inhibitors (NRTIs)

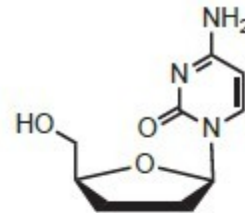
- Act as irreversible, competitive inhibitors for the HIV RT.



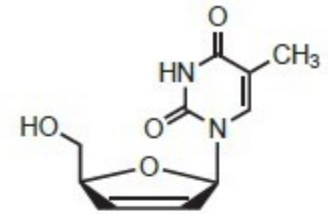
Zidovudine(AZT)



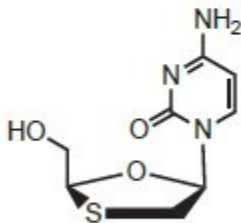
Didanosine(ddI)



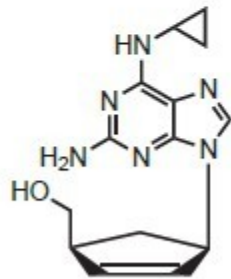
Zalcitabine
(2',3'-Dideoxycytidine, ddC)



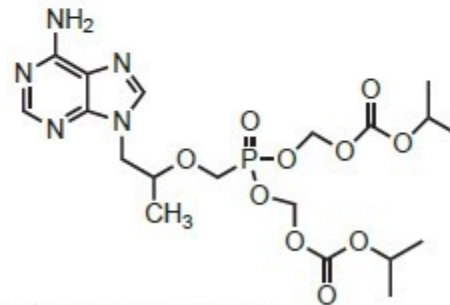
Stavudine
(2',3'-Dideoxy-2',3'-didehydro
thymidine, D4T)



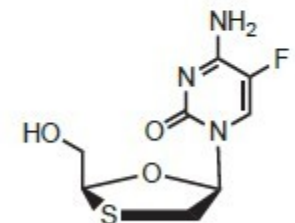
Lamivudine
(2'-Deoxy-3'-thiacytidine, 3TC)



Abacavir (ABC)



Tenofovir disoproxil



Emtricitabine

- Zidovudine** was the first antiviral agent shown to have beneficial effect against HIV infection.
- After prolonged use, AZT-resistant strains rapidly appeared which limited the effect of AZT.
 - It is an analog of thymidine in which the azido group is substituted at the 3 carbon atom of the dideoxyribose moiety.
 - **Stavudine** differs in structure from thymidine by replacement of the 3'hydroxyl group with a hydrogen atom and a double bond in the 2'and 3' positions of deoxyribose ring.

Lamivudine is an analog of 2'-deoxycytidine in which the 3'-methylene group of deoxyribose ring is replaced with a sulfur atom.

Didanosine is a purine dideoxynucleoside which is an analog of inosine .

MOA

They are first bioactivated to triphosphate by cellular kinase.

These phosphates act as:

- Competitive inhibitor of viral reverse transcriptase.
- Chain terminators.

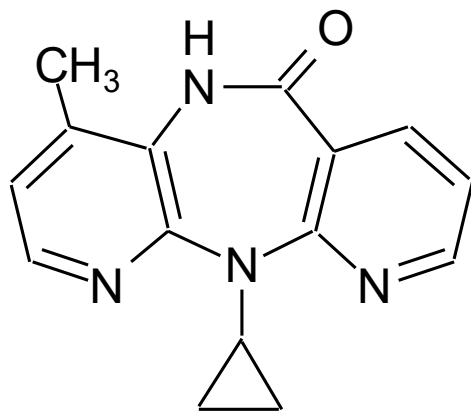
They are incorporated into viral DNA and viral DNA synthesis which is terminated because these drugs lack a 3'–OH group, consequently no more phosphodiester bonds could be formed.

Selectivity of Action

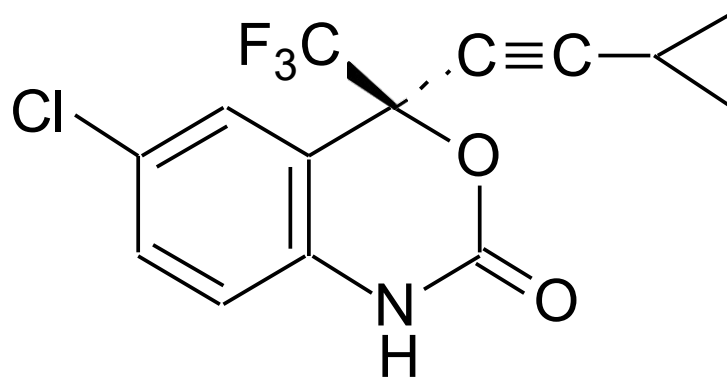
These drugs are less susceptible to mammalian DNA polymerase. Viral reverse transcriptase is 100 times more susceptible to inhibition by zidovudine triphosphate than host cellular DNA polymerase.

Non Nucleoside Analogues Reverse Transcriptase Inhibitors (NNRTI)

- FDA has recently approved several NNRTI.
- They should not be used as monotherapy but they are used with NRTI to obtain synergistic activity in decreasing viral load and increasing CD4+ cell counts.



Nevirapine



Efa
vire
nz

MOA

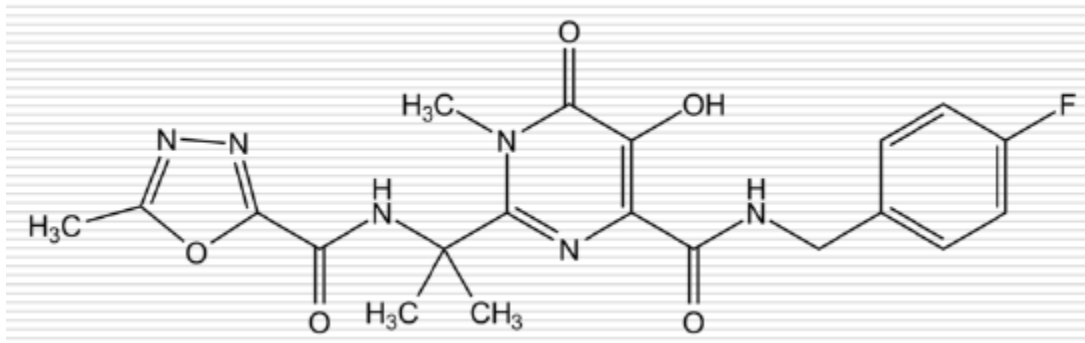
These drugs noncompetitively inhibit viral RT, presumably by binding to a site other than the nucleoside and template binding sites (allosteric site).

This interaction between the NNRTI and the HIV-1 RT produces a conformational change that results in the inactivation of the viral RT.

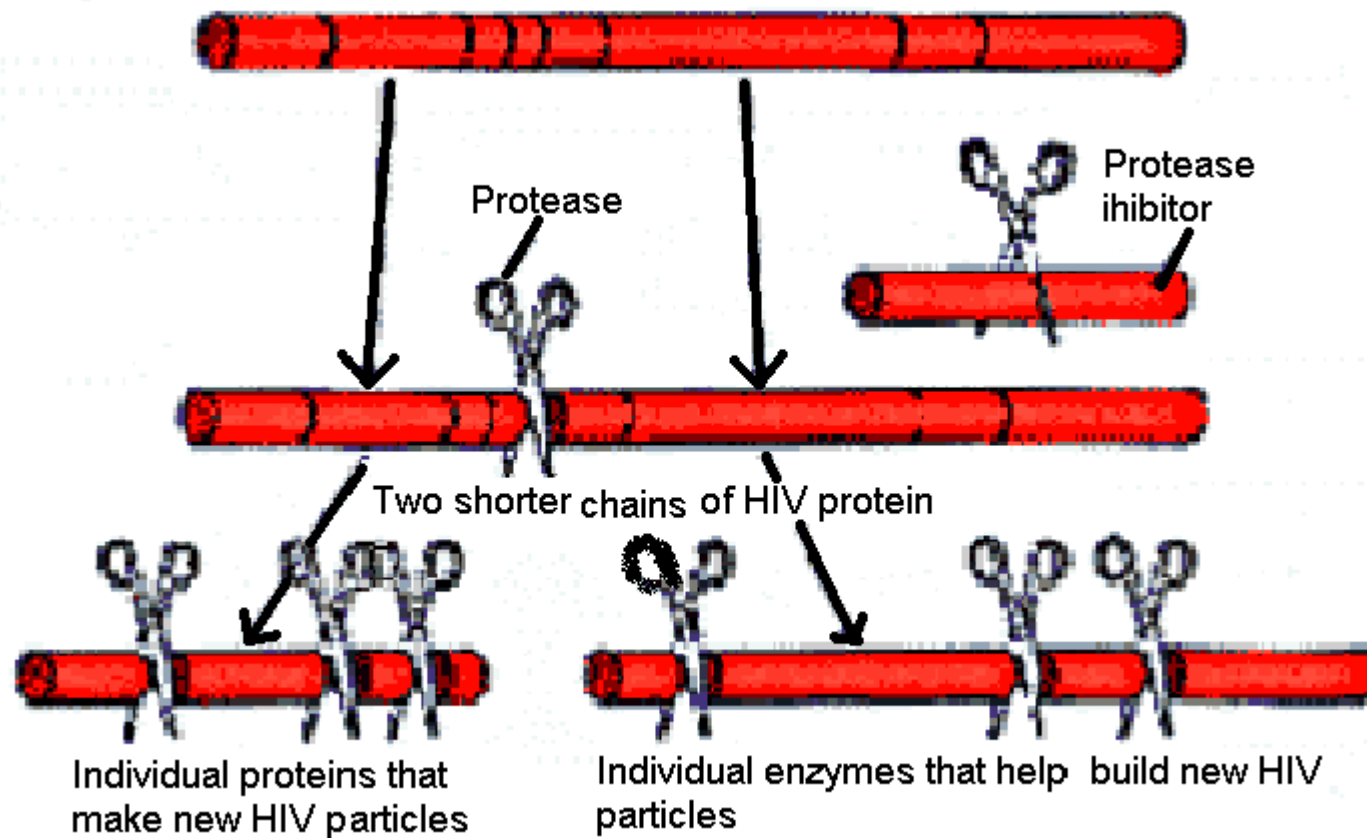
Toxicity includes rash and CNS toxicity as dizziness, insomnia and impaired concentration.

Viral Integrase Inhibitors

- Raltegravir is approved only for use only in individuals whose infection has proven resistant to other HAART drugs.
- As with any HAART medication, raltegravir is unlikely to show durability if used as monotherapy.
- Raltegravir is taken orally twice daily.

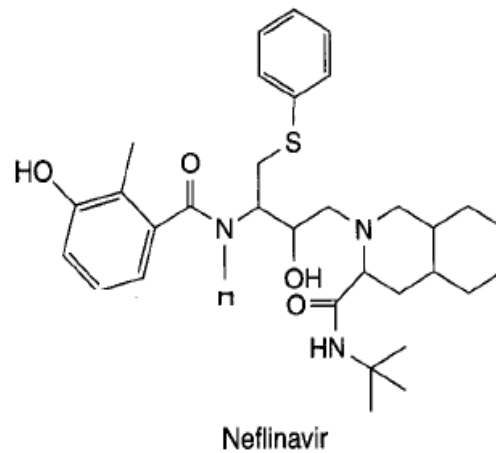
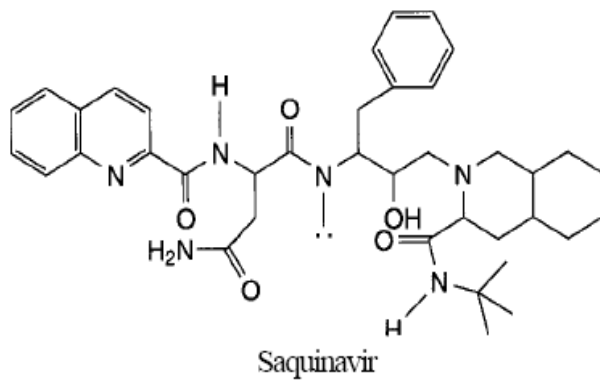
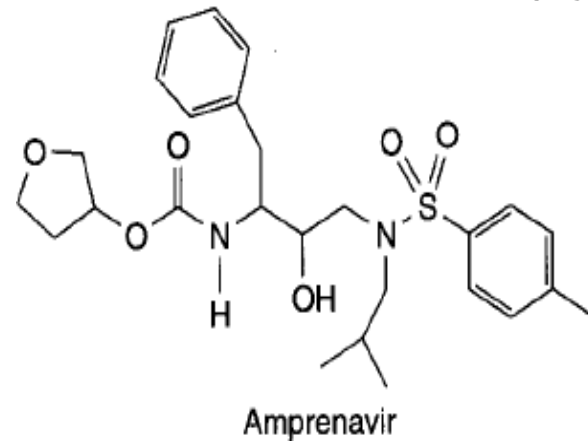
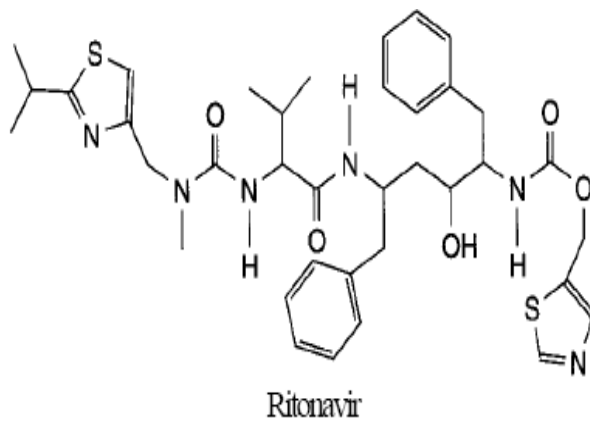
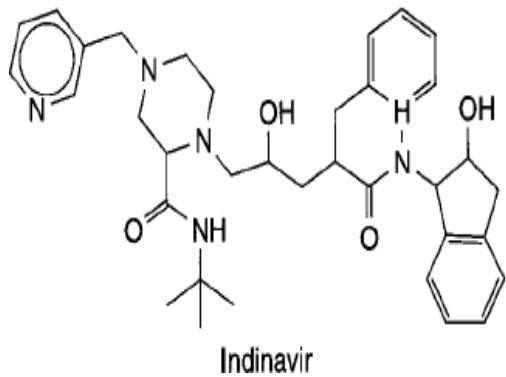


Protease Inhibitors

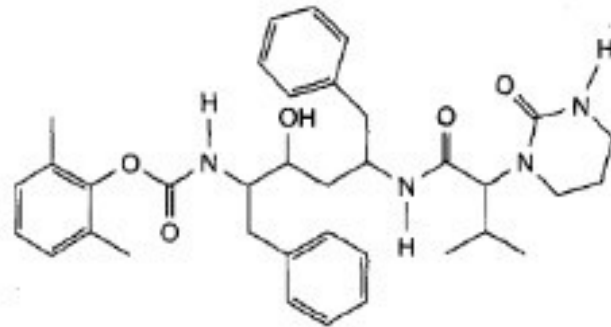


- Inhibitors of this viral protease can be used to fight HIV infection.
- By blocking the ability of protease to cleave the viral polypeptide into functional enzymes, protease inhibitors interfere with continued infection.
- Most current protease inhibitors are complex peptidomimetic compounds with poor aqueous solubility, low bioavailability and short plasma half-lives.
- The complexity of these agents not only contributes to their high cost but also increases the potential for unwanted drug interactions.
- Four protease inhibitors, saquinavir (Invirase®, Hoffman-LaRoche), zidovudine (Norvir®, Abbott), indinavir (Crixivan®, Merck) and nelfinavir (Viracept®, Agouron) have already been approved.

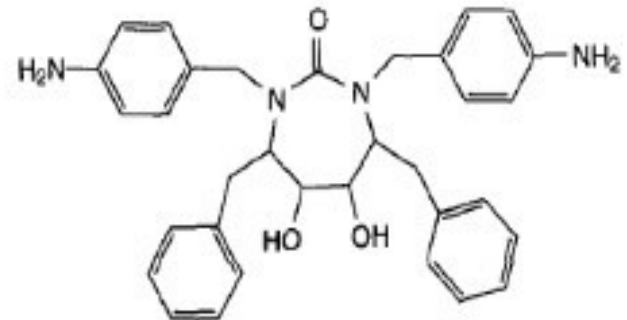
First generation Pis



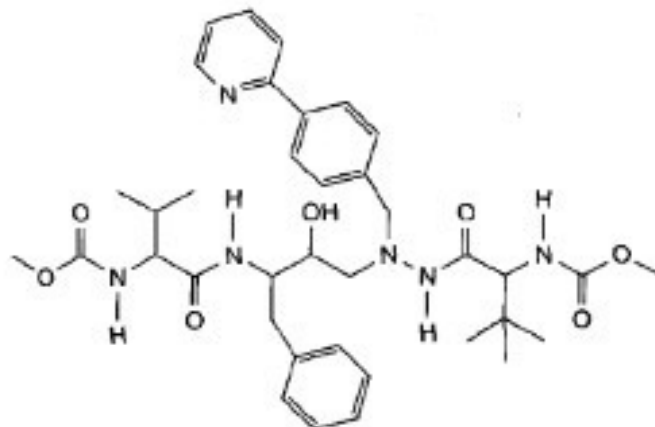
Newer protease inhibitors



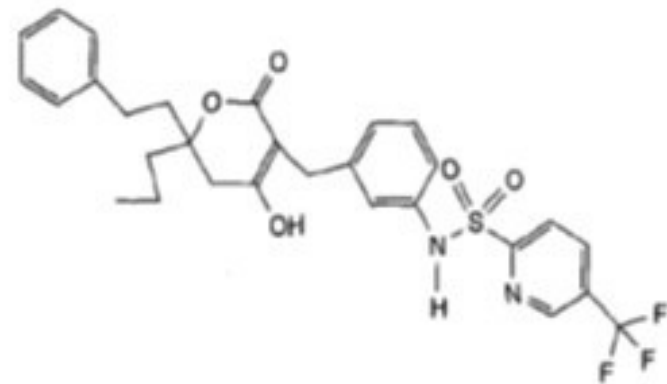
Lopinavir



Mozenavir



Atazanavir



Tipranavir

The aim of anti-HIV therapy has now shifted from simply delaying the progression of the disease to finding a permanent cure.

We have now entered the area of highly active anti-retroviral therapy (HAART).

The current trend is to give a potent combination of agents HAART right from the start when treatment is indicated.

- The most popular combination is AZT and lamivudine plus a protease inhibitor.
- Lamivudine has greater anti-retroviral activity than AZT alone and is active against many AZT-resistant strains without significant increase in toxicity.
- Among protease inhibitors, indinavir (IDV) is more potent than saquinavir and appears to have fewer drug interactions and short-term adverse effects than ritonavir.